

MAT 3150 — Homework 3

Problem 1 Practice: Triangle Inequality Tricks

Tools.

- Triangle inequality: $|a + b| \leq |a| + |b|$ for all $a, b \in \mathbb{R}$.
- $|-a| = |a|$.
- For $c \geq 0$: $|a| \leq c \iff -c \leq a \leq c$.

Worked example

Prove: for all $x, y \in \mathbb{R}$, $|x| - |y| \leq |x + y|$.

Trick: write x as a sum that contains $x + y$:

$$x = (x + y) + (-y).$$

Take absolute values and apply the triangle inequality:

$$|x| = |(x + y) + (-y)| \leq |x + y| + |-y| = |x + y| + |y|.$$

Subtract $|y|$ from both sides:

$$|x| - |y| \leq |x + y|. \quad \checkmark$$

Practice A

Prove: $|y| - |x| \leq |x + y|$.

Hint: same idea, but write y as a sum that contains $x + y$. What do you add to $x + y$ to get back to y ?

Practice B

Prove: $||x| - |y|| \leq |x + y|$.

Hint: let $a = |x| - |y|$ and $c = |x + y|$. What do the worked example and Practice A tell you about a and about $-a$?

Practice C

Prove: for all $x, y, z \in \mathbb{R}$,

$$|x - z| \leq |x - y| + |y - z|.$$

Hint: what can you add to *and* subtract from $x - z$ so it splits into the two pieces on the right?

The pattern

In every one of these, you **rewrite a number as a sum of two pieces**, where one piece is the quantity you want on the right-hand side. Then apply $|a + b| \leq |a| + |b|$ and rearrange. In HW3 Problem 1 that piece is $x - y$; here it is $x + y$.